

What a Census Can and Cannot Do

The one thing nothing else can do, and what it cost to keep doing it

84-355 Democracy's Data

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There are two ways to learn about a country. **Eliciting** asks some people questions; **enumerating** tries to count everyone. This chapter is about the second, and about the price the United States agreed to pay in 2020 in order to keep doing it. The surveys chapter takes up the first, six sessions from now — and the boundary between the two turns out not to run where you would draw it.

A census is the only instrument in this book that is required by the Constitution. Article I orders an “actual Enumeration” every ten years, for one purpose — dividing the House among the states. Everything else the census is used for, which is nearly everything, is a by-product of that obligation.

01 What only an enumeration can do

Here is what the 2020 census resolves to in one state.

Quantity	Value
Blocks in Georgia	232,717
Blocks with nobody in them	67,384
Blocks with at least one person	165,333
Median population of a populated block	26
Blocks containing exactly one person	2,457
Blocks containing fewer than ten people	33,374
Share of Georgians living in blocks of fewer than ten	1.7%
Total population, summed from blocks	10,711,908

232,717 blocks in Georgia. The median populated one holds 26 people. **2,457 of them contain exactly one person.**

No sample can do this, and the reason is arithmetic rather than effort: a survey of sixty thousand people, spread across the country, would place roughly nobody on any given block. The surveys chapter puts a figure to it later in the term. It is not a matter of funding a bigger survey. There is no sample size at which a sample becomes an enumeration.

That capability is load-bearing for most of this book. Every district line is drawn by assembling blocks. Every rate with a population on the bottom of the fraction needs

a count of the people in a place. That bottom number is called the **denominator**, and turnout, registration and stops per resident all need one, and that count comes from here or it comes from nowhere.

02 Three of the Bureau's 134 instruments get called "the census"

Start with what the agency actually runs. Its own Survey Explorer lists **134 surveys and censuses**. The American Housing Survey, the Commodity Flow Survey, the Census of Jails, the Quarterly Survey of Public Pensions, the National Crime Victimization Survey, and a hundred and thirty others. The Bureau is a general-purpose statistical agency that also happens to conduct the enumeration.

Three of those 134 get the rest of this chapter, and the reason is not that the other 131 are minor. It is that these three are the ones routinely mistaken for each other. Each answers some version of *how many people are in this place*; each publishes a figure formatted like a population; and each carries the Bureau's name. Nobody quotes the Commodity Flow Survey and means a headcount – that confusion never happens, because nothing about the two instruments looks alike.

These three also carry the rest of this book. A district, a turnout rate, a stops-per-resident figure: each needs a number of people in a place, and that number comes from the decennial census, the American Community Survey, or the population estimates. The other 131 programs measure other things, and a reader who reaches for the wrong one of *those* has usually made a visible mistake rather than a silent one.

So: almost everyone who says "the census says" is quoting one of three instruments, and they behave differently.

The question	Decennial census	American Community Survey	Population estimates
How many people live on this block?	Yes – this is the one thing only it can do	No – the sample is far too thin	No
How many people moved from Ohio to Georgia last year?	No – it does not ask	Yes, with a margin often larger than the estimate	No
How many people live in this county today?	No – it is ten years old the day after it is taken	Yes, as a five-year average centred on the past	Yes – this is what it is for
What is the population of the United States, exactly?	As of one day, every ten years, subject to undercount	No – it is a sample	No – it is a model carried forward from the last census
How many people here speak Spanish and little English?	No – the short form does not ask	Yes – this is what it is for	No
Is this number exact?	No – noise is added below the state level	No – and it prints its own margin of error	No – it is a projection and is revised

The **decennial census** is the enumeration: everyone, one day, every ten years. The **pop-**

ulation estimates carry it forward with births, deaths and migration – a model, revised annually, and the source of nearly every “current population of X” you will read.

And the **American Community Survey** is a survey. It samples about three and a half million addresses a year, and it produces estimates rather than counts. It prints a margin of error next to every one of them: a stated range for how far off that estimate could be. It is published by the same agency, in the same style, on the same website, and it is quoted as though it were a count constantly.

03 A survey in the counting part, which is not a mistake

That should raise an objection, and it is worth answering now rather than later. If the ACS asks some people questions, it is doing the thing this part just called *eliciting* – so why is it here, in a part about counting, instead of with the surveys?

Because answering it is not optional. Refusal to answer the ACS is punishable under 13 U.S.C. § 221, the same section that compels the decennial count, and the envelope says so: *your response is required by law*. Every other survey in this book is voluntary. The American National Election Studies asks people whether they would like to take part and most say no; nobody has ever been fined for declining the General Social Survey.

So the two words in this part’s title are doing different work than they look like they are.

This part is not about counting as a method. It is about data somebody was ordered to produce, and the census, the ACS and the estimates all follow from the same obligation. The part that comes later and *is* about method is Asking People. Nothing there had to exist. Every number in it is there because a researcher chose to ask, and found the money to pay for it.

Keep the pair in view, because the contrast is the sharpest one this book has. **Two samples, drawn from the same country in the same decade, asking people questions, and one of them can fine you for hanging up.** What each can be made to say follows from that and almost nothing else.

Here is what that costs. These are the Bureau’s own published state-to-state migration flows:

Quantity	Value
Published state-to-state flows	2,652
Flows whose margin of error exceeds the estimate	654
Share of flows whose margin exceeds the estimate	27.6%
Flows distinguishable from zero	1,714
Median published estimate	970
Median published margin of error	700

654 of 2,652 published flows, 27.6% of them, have a margin of error larger than the estimate itself. The median published flow is 970 people, with a median margin of 700.

Nothing is hidden here. The Bureau prints the margin beside the estimate in every table. But a number that arrives from the Census Bureau, formatted like a count, is read as a

count, and a quarter of these are not distinguishable from a rather different number. The migration chapter is what happens when you take those margins seriously.

Before you read on, commit to a view. A news story reports that 4,200 people moved from Ohio to Georgia last year, citing the Census Bureau. **What is the first question you ask**, and where in this chapter would you look for the answer?

04 One state, two official populations

Population	Value	What it is for
Resident population, 2020 census	10,711,908	Redistricting, funding formulas, every rate with a population denominator
Apportionment population, 2020 census	10,725,274	Dividing the 435 House seats among the states
Difference	13,366	Federal employees and their dependents stationed overseas, allocated to a home state for apportionment only

Georgia's 2020 population is 10,711,908. It is also 10,725,274. Both figures are official and both come from the 2020 census, and they differ by 13,366 people. Those are federal employees and their dependents stationed overseas, assigned to a home state for the purpose of dividing House seats, and not residents of it for any other purpose.

Neither number is wrong. They are answers to two different legal questions. A person who takes the apportionment figure as the population, or divides by it to get a rate, has silently changed the question. This is the book's recurring problem in its most official possible form: **a denominator nobody states**, published by the agency whose job is stating things.

05 The published count is not the collected count

In 2020 the Census Bureau did something it had never done before at this scale. A file that can describe a block of 26 people by race and age can also identify them, and 2,457 Georgia blocks contain exactly one person. So the Bureau applied a formal privacy system that adds noise to the published counts below the state level.

The counts you have been reading in this chapter are not, exactly, the counts the Bureau collected.

Two things about that are worth verifying rather than taking on faith, and both can be checked from the file.

First, the state total is exact. Sum all 232,717 noised Georgia block counts and you get 10,711,908 — the published resident population, to the person. The top of the hierarchy is held fixed and the noise beneath it is constrained to sum back to it.

Second, and this is the part that matters: you cannot find the noise.

Consistency check	Violations	Blocks checked
Blocks where voting-age population exceeds total population	0	232,717
Blocks where Black population exceeds total population	0	232,717
Blocks where Black voting-age population exceeds Black population	0	232,717
Blocks with a negative count in any field	0	232,717

Every internal check passes. Not one block has more voting-age people than people. Not one has more Black residents than residents. Not one negative count, in 232,717 blocks. A reader who suspected the file had been altered and went looking for impossibilities would find none and would conclude, reasonably and wrongly, that nothing had been done to it.

The noise is real, it is documented, and it is invisible from inside the data. **The only reason anyone knows the published counts differ from the collected counts is that the Bureau said so.** No audit of the file recovers that fact.

This is the strongest possible version of a lesson that runs through the whole book. A file's trustworthiness is not a property you can test for by staring at the numbers. It is a property of the institution that produced it and of what that institution chose to tell you.

A second prediction. The trade-off is real: more noise means better privacy and worse small-area counts. **Who is hurt most by the noise, and who is hurt most by its absence?** Name a specific use from this book for each.

06 What arrives

The P.L. 94-171 file arrives as four pipe-delimited files with no header row. A block's population is in segment 1; who those people are by race is further along the same line; the block's NAME and location are in a fourth file, joined on a record number that is unique within a state and means nothing outside it.

Three real Georgia blocks, after the join, as the derived extract stores them:

```

GE0ID20  pop  black_alone  black_any  vap  black_any_vap
130670306032011  15  1  4  12  1
131590105031067  25  3  3  16  0
130771703031017  36  0  2  34  0

```

GE0ID20 is fifteen characters and must be read as text. As a number it loses its leading zero and stops being a Georgia block at all.

Every count above is a published count. It is not necessarily the count that was collected -- see the chapter.

07 What this chapter cannot tell you

The block statistics are Georgia's. Block counts, sizes and the share of one-person blocks differ by state, and a rural state would look different from this one. The arithmetic that makes small areas impossible for a sample does not differ.

The other 131 programs are not covered. Three instruments were chosen because they are the three that get confused with each other, not because they are the three that matter. If a question here is about housing conditions, commuting, crime victimisation, business receipts or public pensions, the answer is on the Bureau's list and not in this chapter. The Survey Explorer is the place to find which survey asks it. That list also changes: surveys are added, terminated and re-engineered, and the count of 134 is what the tool showed when this chapter was written.

Nothing here measures the undercount. A census attempts to count everyone and does not succeed; coverage error is estimated afterward by a separate survey, it has never been zero, and it has never fallen equally on everyone. This chapter is about what the published file is, not about how close it came. The apportionment chapter is where that bites, because a seat can turn on fewer people than the coverage error.

This chapter also does not say how *much* noise was added, or where. The size of the adjustment at any particular block is not published, which is the point of it. And the Bureau's own accuracy metrics are a large technical literature this chapter does not summarise. What is verified here is narrower and, for a reader of files, more useful: the total is exact, and the consistency checks cannot detect anything.

Finally, the categories. Every count in this chapter is a count of people in boxes that the Office of Management and Budget defines and periodically revises. The census does not discover how many Black Georgians there are; it counts the people who answered a particular question a particular way in 2020. That is the census chapter's subject, and it is why "how many" is always a shorter question than it looks.

08 Sources

- U.S. Census Bureau. *2020 Census P.L. 94-171 Redistricting Data Summary File*, Georgia. Block counts read from the copy committed for the areal-units chapter.
- 13 U.S.C. § 221, which makes refusal to answer either the decennial census or the American Community Survey an offence — the obligation that puts a sample survey in this part rather than with the surveys. <https://www.law.cornell.edu/uscode/text/13/221>
- U.S. Census Bureau. *Census Survey Explorer*, accessed August 2026. The full inventory of the Bureau's surveys and censuses, filterable by geography, frequency and topic. <https://www.census.gov/data/data-tools/survey-explorer/>
- U.S. Census Bureau. *2020 Census Apportionment Results*, 26 April 2021. <https://www.census.gov/data/tables/2020/dec/2020-apportionment-data.html>
- U.S. Census Bureau. *American Community Survey*, state-to-state migration flows, via this book's migration chapter.
- U.S. Constitution, Article I, Section 2, and the Census Act, 13 U.S.C.
- boyd, danah and Jayshree Sarathy (2022). "Differential Perspectives: Epistemic Disconnects Surrounding the U.S. Census Bureau's Use of Differential Privacy." *Harvard Data Science Review*.

- Ruggles, Steven and David Van Riper (2022). "The Role of Chance in the Effective Use of Differential Privacy." *Population Research and Policy Review* 41: 1-14.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`acs.csv`, `consistency.csv`, `granularity.csv`, `instruments.csv`, `populations.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `instruments.csv` puts the same questions to three instruments and records which of them can answer. Find a question only one can answer. That answer is the reason the whole enterprise still exists.
- `populations.csv` gives one state two official populations, each with a `what_it_is_for`. Decide which belongs in a turnout rate, then work out what goes wrong if you use the other.
- `consistency.csv` lists checks that came back clean. Pick one and describe the real data error it would have caught, in a sentence someone could act on.
- `granularity.csv` counts the blocks in a state. Divide the state's population by that count. What is a typical block, and what can you not say about the people in one?

Watched

- U.S. Census Bureau, "Census 101: An Introduction to the Census Bureau." The Bureau's own account of what it is and what it collects — worth watching before this chapter for the version the agency tells about itself, and again after it for what that version leaves out. <https://youtu.be/ztmKRYnwq5A>